

ES 211 - Thermodynamics

Project

Enhancing Performance and Efficiency of an Ideal Rankine Cycle

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Problem Statement

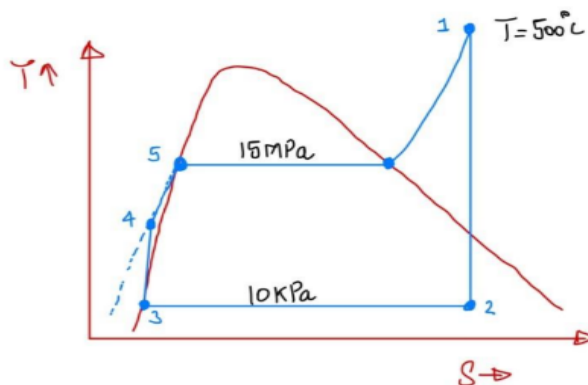
In this project, the objective is to enhance the performance of a power plant currently operating on a basic ideal Rankine cycle. The existing cycle exhibits a thermal efficiency of 41% and a steam quality of 76% at the condenser inlet. The goal is to modify the ideal Rankine cycle to achieve a thermal efficiency exceeding 46% and to raise the steam quality (x) to a level above 85%. Additionally, it is essential to maintain the boiler pressure (P_b) and condenser pressure (P_c) at 15 MPa and 10 kPa, respectively, while ensuring that the turbine temperature does not exceed 500 °C.

To accomplish this, the key parameters to be adjusted are the boiler pressure (P_b) and condenser pressure (P_c). These variables can be varied within the specified ranges of $12 \text{ MPa} < P_b < 15 \text{ MPa}$ and $5 \text{ kPa} < P_c < 10 \text{ kPa}$. The objective is to investigate how changes in these pressures impact the thermal efficiency and net work output of the modified ideal Rankine cycle.

The project involves:

- (a) Modifying the ideal Rankine cycle to achieve a thermal efficiency greater than 46% and a steam quality exceeding 85%.
- (b) Conducting a parametric analysis by varying the boiler pressure (P_b) and condenser pressure (P_c) within the specified ranges and creating a plot to visualize the impact of these pressure changes on the thermal efficiency and net work output of the modified cycle.

This project aims to optimize the Rankine cycle parameters to meet the specified efficiency and steam quality targets, contributing to the overall efficiency and performance improvement of the power plant.



Rankine Cycle

The Rankine cycle is a thermodynamic cycle commonly used in steam power plants to generate electricity. It is named after the Scottish engineer William Rankine who developed the concept. The Rankine cycle is a theoretical representation of the operation of a steam power system and consists of four main processes: compression, heating, expansion, and cooling.

Here's a brief overview of the four stages of the Ideal Rankine cycle:

- ➔ Isentropic Compression (Process 3-4): The working fluid, often water or steam, is compressed in a pump from low pressure to high pressure. This process is usually isentropic, meaning it occurs with constant entropy.
- ➔ Isobaric Heating (Process 4-1): The high-pressure fluid is then heated in a boiler at constant pressure. This heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in volume and temperature.
- ➔ Isentropic Expansion (Process 1-2): The high-pressure, high-temperature vapor expands through a turbine, converting some of its internal energy into mechanical work. This process is typically isentropic.
- ➔ Isobaric Cooling (Process 2-3): The low-pressure vapor is then condensed back into a liquid in a condenser, releasing heat to the surroundings. The fluid is then pumped back to the high-pressure state, completing the cycle.

The Rankine cycle is a practical model for understanding and analyzing the thermodynamic processes in steam power plants. While idealized, it provides a foundation for the design and operation of power generation systems. Real-world power plants may incorporate additional components and considerations, such as **reheating** and **regeneration**, to improve efficiency and performance.

Reheat and Regeneration

Reheat and regeneration are two thermodynamic concepts commonly employed in steam power plants, specifically in the context of the Rankine cycle, to enhance overall efficiency and performance.

Reheat:

- Process Description: In a reheat cycle, the steam is extracted from the turbine after partial expansion, and it is then sent back to the boiler for reheating before entering another stage of expansion in the turbine.
- Purpose: Reheating the steam allows it to regain some of its lost temperature and pressure, enabling it to undergo further expansion in the turbine. This increases

the overall work output from the turbine and, consequently, the overall efficiency of the Rankine cycle.

- Advantages: Reheat reduces the moisture content (increases steam quality) and temperature drop of the steam during its expansion in the turbine, resulting in a more efficient and effective power generation cycle.

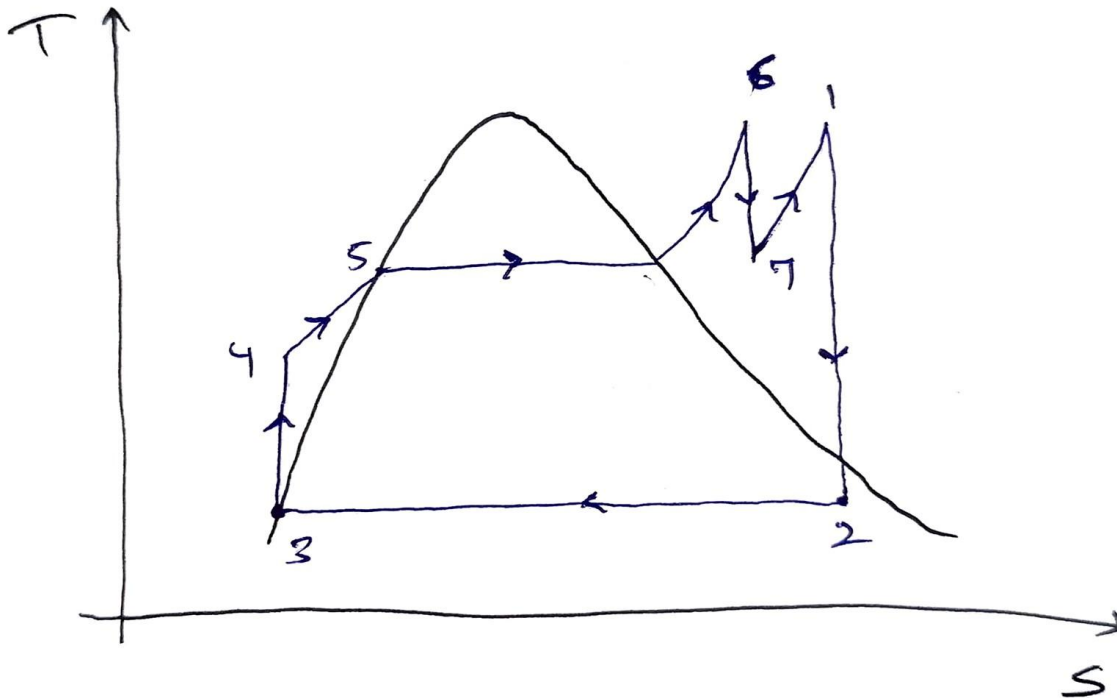
Regeneration:

- Process Description: Regeneration involves the extraction of steam from the turbine at an intermediate stage and using it to preheat the feedwater before it enters the boiler.
- Purpose: By preheating the feedwater using extracted steam, less heat needs to be added in the boiler to reach the desired operating conditions. This process reduces the overall amount of fuel required for steam generation and improves the cycle efficiency.
- Advantages: Regeneration helps to utilize the waste heat from the turbine more effectively, enhancing the thermal efficiency of the Rankine cycle. It also contributes to a reduction in fuel consumption and operational costs.

Both reheat and regeneration are methods employed to address limitations and improve the efficiency of the Rankine cycle, which is the basis for many steam power plants. These techniques allow for better utilization of the available heat and enhance the power generation process by mitigating issues such as moisture content in the turbine and minimizing the heat input required in the boiler.

1. We first applied one reheater and checked the efficiency and quality.

The Rankine cycle with one reheater is an extension of the basic Rankine cycle, incorporating an additional component for reheating the steam. Here's a brief overview of the processes that occur in the Rankine cycle with one reheater:



→ Isentropic Compression (Process 3-4):

- ◆ The working fluid, often water or steam, is compressed in a pump from low pressure to high pressure.
- ◆ This process is usually isentropic, meaning it occurs with constant entropy.

→ Isobaric Heating in Boiler (Process 4-6):

- ◆ The high-pressure fluid is then heated in the boiler at constant pressure.
- ◆ Heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in temperature and volume.
- ◆ Steam leaves the boiler at a high temperature and pressure.

→ Isentropic Expansion in High-Pressure Turbine (Process 6-7):

- ◆ The high-pressure, high-temperature steam expands through the high-pressure turbine, converting some of its internal energy into mechanical work.
- ◆ The steam exits the high-pressure turbine at an intermediate pressure and temperature.
- Reheat (Process 7-1):
 - ◆ The intermediate-pressure steam is extracted from the high-pressure turbine and sent to the reheater, where it is heated at constant pressure.
 - ◆ Reheating the steam improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Low-Pressure Turbine (Process 1-2):
 - ◆ The reheated steam expands through the low-pressure turbine, further converting its internal energy into mechanical work.
 - ◆ The steam exits the low-pressure turbine and is directed to the condenser.
- Isobaric Cooling in Condenser (Process 2-3):
 - ◆ The low-pressure steam is condensed back into a liquid in the condenser, releasing heat to the surroundings.
 - ◆ The condensed liquid is pumped back to the high-pressure state, completing the cycle.

The inclusion of a reheater allows the Rankine cycle to maintain high temperatures during the expansion phases, reducing moisture content in the turbine and improving overall cycle efficiency. The specific details of the cycle, such as pressures, temperatures, and the number of turbine stages, can vary depending on the design and requirements of the power plant.

Condenser as control volume,

$$\dot{Q}_{out} = h_2 - h_3$$

Boiler as CV, $\dot{Q}_b = h_6 - h_4$

Reheater as CV: $\dot{Q}_r = (h_7 - h_1)$

$$\dot{Q}_{in} = h_6 - h_4 + (h_7 - h_1)$$

$$\begin{aligned} Efficiency(\eta) &= 1 - \frac{Q_{out \{2 \rightarrow 3\}}}{Q_{in \{6 \rightarrow 4, 1 \rightarrow 7\}}} \\ &= 1 - \frac{h_2 - h_3}{(h_6 - h_4) + (h_1 - h_4)} \end{aligned}$$

$$\begin{aligned} \Rightarrow Net \ work \ output &= Q_{in} - Q_{out} \\ &= (h_6 - h_4 + h_7 - h_1 + h_3 - h_2) \end{aligned}$$

Matlab code for 1 reheater

```
clc;
clear all;
% 1 reheater
w = Solution('liquidvapor.cti', 'water');
P3 = 10e+3;
P2 = P3;
P4 = 15e+6;
P5 = P4;
P6 = P4;
T6 = 500 + 273.15;
set(w, 'P', P6, 'T', T6);
h6 = enthalpy_mass(w);
s6 = entropy_mass(w);
s7 = s6;
P7 = 3e+6;
setState_SP(w, [s7, P7]);
h7 = enthalpy_mass(w);
P1 = P7;
T1 = T6;
set(w, 'P', P1, 'T', T1);
h1 = enthalpy_mass(w);
s1 = entropy_mass(w);
s2 = s1;
setState_SP(w, [s2, P2]);
```

```

h2 = enthalpy_mass(w);
setState_Psat(w, [P3, 0]);
h3 = enthalpy_mass(w);
s3 = entropy_mass(w);
s4 = s3;
setState_SP(w, [s4, P4]);
h4 = enthalpy_mass(w);
% Calculating the efficiency
efficiency = 1 - (h2 - h3)/(h6 - h4 + h1 - h7);
% quality at state 2
setState_Psat(w, [P2, 0.0]);
sf = entropy_mass(w);
setState_Psat(w, [P2, 1.0]);
sg = entropy_mass(w);
sfg = sg - sf;
x = (s2 - sf)/sfg;
%% Printing the final results
fprintf("The efficiency of this cycle is %.4f %%\n", efficiency*100);
fprintf("The quality at condenser inlet is %.4f %%\n", x*100);

```

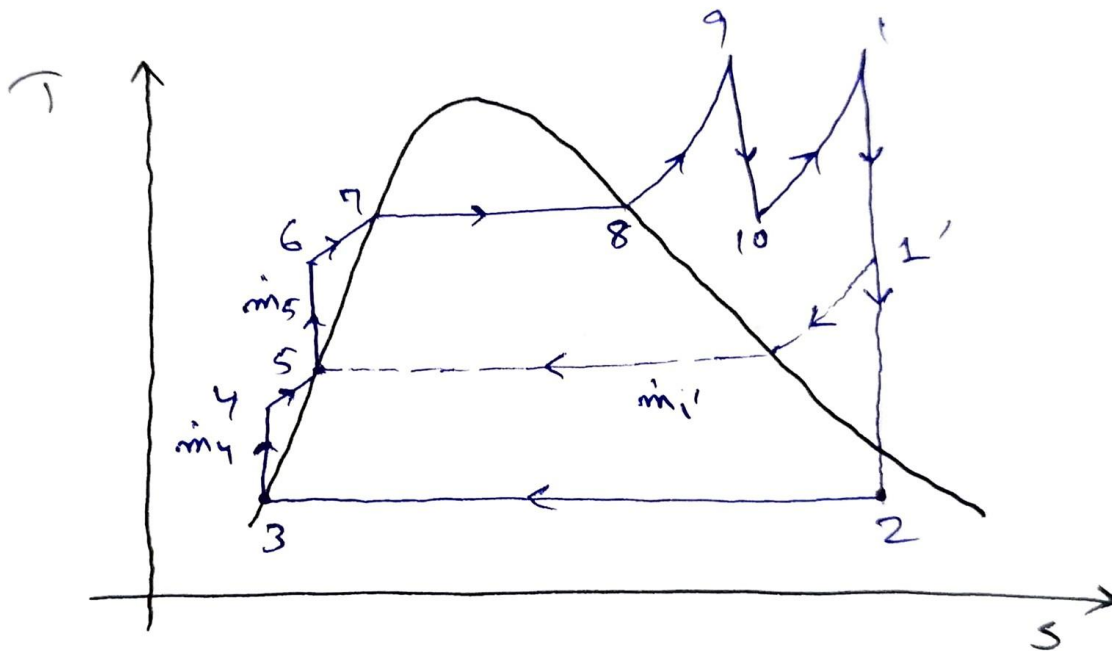
We found -

The efficiency of this cycle is 42.8025 %

The quality at condenser inlet is 87.7788 %

2. Modified Ideal Rankine Cycle with 1 reheater and 1 regeneration.

The Rankine cycle with one reheater and one regeneration is an advanced configuration that includes both a reheating process and a regeneration process. Here's a brief overview of the processes that occur in the Rankine cycle with one reheater and one regeneration:



→ Isentropic Compression (Process 3-6):

- ◆ The working fluid, often water or steam, is compressed in a pump from low pressure to high pressure.
- ◆ This process is usually isentropic, occurring with constant entropy.

→ Isobaric Heating in Boiler (Process 6-9):

- ◆ The high-pressure fluid is then heated in the boiler at constant pressure.
- ◆ Heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in temperature and volume.
- ◆ Steam leaves the boiler at a high temperature and pressure.

→ Isentropic Expansion in High-Pressure Turbine (Process 9-10):

- ◆ The high-pressure, high-temperature steam expands through the high-pressure turbine, converting some of its internal energy into mechanical work.
- ◆ The steam exits the high-pressure turbine at an intermediate pressure and temperature.
- Reheat (Process 10-1):
 - ◆ The intermediate-pressure steam is extracted from the high-pressure turbine and sent to the reheater, where it is heated at constant pressure.
 - ◆ Reheating the steam improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Low-Pressure Turbine (Process 1-2):
 - ◆ The reheated steam expands through the low-pressure turbine, further converting its internal energy into mechanical work.
 - ◆ The steam exits the low-pressure turbine and is directed to the next stage of the cycle.
- Regeneration (Process 1'-5):
 - ◆ Some of the steam from the low-pressure turbine is extracted and sent to a feedwater heater.
 - ◆ The extracted steam transfers heat to the feedwater, increasing its temperature before it enters the boiler.
 - ◆ This process improves the overall efficiency of the cycle by utilizing the waste heat to preheat the feedwater.
- Isobaric Cooling in Condenser (Process 2-3):
 - ◆ The remaining low-pressure steam is condensed back into a liquid in the condenser, releasing heat to the surroundings.
 - ◆ The condensed liquid is pumped back to the high-pressure state, completing the cycle.

The combination of reheating and regeneration in the Rankine cycle contributes to increased efficiency by reducing moisture content in the turbine and utilizing waste heat to preheat the feedwater, respectively. The specific details of the cycle, such as pressures, temperatures, and the number of turbine stages, can vary depending on the design and requirements of the power plant.

Calculations

At State (5), Open feedwater heater as CV:

$$\text{mass balance} \Rightarrow \dot{m}_4 + \dot{m}_1 = \dot{m}_5 \quad \dots (i)$$

$$\text{energy balance} \Rightarrow \dot{m}_4 h_4 + \dot{m}_1 h_1 = \dot{m}_5 h_5 \quad \dots (ii)$$

$$\dot{m}_4 h_4 + (\dot{m}_5 - \dot{m}_4) h_i = \dot{m}_5 h_5$$

$$\dot{m}_4 (h_4 - h_1') = \dot{m}_5 (h_5 - h_1)$$

From (i) & (ii)

$$\frac{\dot{m}_4}{\dot{m}_5} = \dot{m}_{4-5} = \frac{h_5 - h_1'}{h_4 - h_1'} \quad \dots (iii)$$

Condenser as CV

$$Q_{out} = \dot{m}_4 (h_2 - h_3)$$

$$= 1 - \dot{m}_4 \left(\frac{h_2 - h_3}{h_9 + h_1 - h_{10} - h_6} \right)$$

From Isobaric heat addition & reheater

$$\dot{Q}_{in} = \dot{m}_5(h_9 - h_6) + \dot{m}_5(h_1 - h_{10})$$

$$\eta = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{\dot{m}_4(h_2 - h_3)}{\dot{m}_5(h_9 - h_6) + \dot{m}_5(h_1 - h_{10})}$$

Using (iii)

$$\eta = 1 - \left(\frac{h_1' - h_5}{h_1' - h_4} \right) \left(\frac{h_2 - h_3}{h_9 - h_6 + h_1 - h_{10}} \right)$$

$$\text{Net work Output} = (\dot{Q}_{in} - \dot{Q}_{out}) \text{ or } (\eta \times \dot{Q}_{in})$$

$$= \dot{m}_5 \left((h_9 - h_6 + h_1 - h_{10}) - \frac{\dot{m}_4(h_2 - h_3)}{\dot{m}_5} \right)$$

Matlab code to calculate efficiency and quality for 1 reheater + 1 regeneration

```
clc;
clear all;
%%
h = Solution('liquidvapor.cti', 'water');
%% State 3
P3 = 10E+3;
setState_Psat(h, [P3, 0]);
h3 = enthalpy_mass(h);
s3 = entropy_mass(h);
%% State 4
P4 = 0.8E+6; % Vary
s4 = s3;
setState_SP(h, [s4, P4]);
h4 = enthalpy_mass(h);
%% State 5
P5 = P4;
```

```

setState_Psat(h,[P5,0]);
h5 = enthalpy_mass(h);
s5 = entropy_mass(h);
%% State 6
P6 = 15E+6;
s6 = s5;
setState_SP(h,[s5,P6]);
h6 = enthalpy_mass(h);
%% State 7
P7 = P6;
setState_Psat(h,[P7,0]);
h7 = enthalpy_mass(h);
s7 = entropy_mass(h);
%% State 8
P8 = P6;
setState_Psat(h,[P8,1]);
h8 = enthalpy_mass(h);
s8 = entropy_mass(h);
%% State 9
P9 = P6;
T9 = 500+273.15;
set(h,'P',P9,'T',T9);
h9 = enthalpy_mass(h);
s9 = entropy_mass(h);
%% State 10
P10 = 3.8E+6; % vary
s10 = s9;
setState_SP(h,[s10,P10]);
h10 = enthalpy_mass(h);
%% State 1
P1 = P10;
T1 = 500+273.15;
set(h,'P',P1,'T',T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);
%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h,[s1_,P1_]);
h1_ = enthalpy_mass(h);
%% State 2
P2 = P3;
s2 = s1;
setState_SP(h,[s2,P2]);
h2 = enthalpy_mass(h);
setState_Psat(h,[P2,0]);
hf = enthalpy_mass(h);
setState_Psat(h,[P2,1]);
hg = enthalpy_mass(h);
hfg = hg-hf;
x = (h2-hf)/hfg;
m4_5 = ((h1_-h5)/(h1_-h4));
qout = m4_5*(h2-h3);
qin = h9-h6+h1-h10;
efficiency = 1-(qout/qin);
%% Printing the final results

```

```
fprintf("The efficiency of this cycle is %.4f %%\n", efficiency*100);  
fprintf("The quality at condenser inlet is %.4f %%\n", x*100);
```

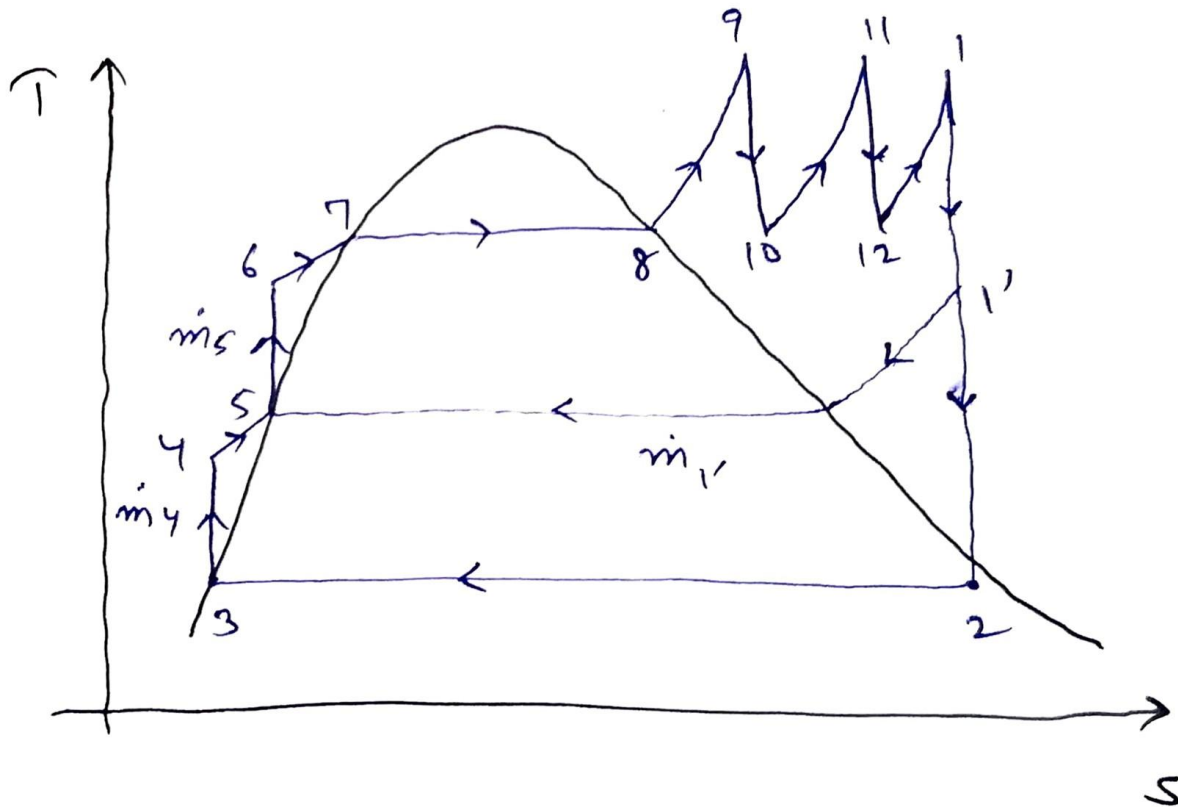
We found-

```
The efficiency of this cycle is 45.6765 %  
The quality at condenser inlet is 86.2073 %
```

The Rankine Cycle, enhanced with a reheater and regeneration, exhibits a steam quality exceeding 85% at the condenser inlet. However, despite these improvements, the cycle falls short of achieving an efficiency surpassing 46%.

3. Modified Ideal Rankine Cycle with 2 reheater and 1 regeneration:

The Rankine cycle with two reheaters and one regeneration is an advanced configuration that combines multiple features to enhance the efficiency of the power generation process. Here's a brief overview of the processes that occur in the Rankine cycle with two reheaters and one regeneration:



→ Isentropic Compression (Process 3-6):

- ◆ The working fluid, typically water or steam, is compressed in a pump from low pressure to high pressure.
- ◆ This process is generally isentropic, occurring with constant entropy.

→ Isobaric Heating in Boiler (Process 3-9):

- ◆ The high-pressure fluid is heated in the first stage of the boiler at constant pressure.
- ◆ Heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in temperature and volume.

- ◆ Steam leaves the first stage of the boiler at a high temperature and pressure.
- Isentropic Expansion in High-Pressure Turbine (Process 9-10):
 - ◆ The high-pressure, high-temperature steam expands through the high-pressure turbine, converting some of its internal energy into mechanical work.
 - ◆ The steam exits the high-pressure turbine at an intermediate pressure and temperature.
- Reheat 1 (Process 10-11):
 - ◆ The intermediate-pressure steam is extracted from the high-pressure turbine and sent to the first reheater, where it is heated at constant pressure.
 - ◆ Reheating the steam improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Intermediate-Pressure Turbine (Process 11-12):
 - ◆ The reheated steam expands through an intermediate-pressure turbine, further converting its internal energy into mechanical work.
 - ◆ The steam exits the intermediate-pressure turbine at a lower pressure and temperature.
- Reheat 2 (Process 12-1):
 - ◆ The low-pressure steam is extracted from the intermediate-pressure turbine and sent to the second reheater, where it undergoes further reheating at constant pressure.
 - ◆ Reheating the steam again improves its quality by increasing its temperature without raising its pressure.
- Regeneration (Process 1'-5):
 - ◆ Some of the steam from the second reheater is extracted and sent to a feedwater heater.
 - ◆ The extracted steam transfers heat to the feedwater, increasing its temperature before it enters the first stage of the boiler.
 - ◆ This process improves the overall efficiency of the cycle by utilizing the waste heat to preheat the feedwater.
- Isentropic Expansion in Low-Pressure Turbine (Process 1-2):
 - ◆ The twice-reheated steam expands through the low-pressure turbine, converting its internal energy into mechanical work.
 - ◆ The steam exits the low-pressure turbine and is directed to the condenser.
- Isobaric Cooling in Condenser (Process 2-3):
 - ◆ The low-pressure steam is condensed back into a liquid in the condenser, releasing heat to the surroundings.

- ◆ The condensed liquid is pumped back to the high-pressure state, completing the cycle.

This configuration incorporates both reheating and regeneration to optimize the Rankine cycle for increased efficiency and improved steam quality. The specific details of the cycle, such as pressures, temperatures, and the number of turbine stages, can vary depending on the design and requirements of the power plant.

Calculations

At state (5), open feedwater heater as C.V :-

$$\text{mass balance} = \dot{m}_4 + \dot{m}_1 = \dot{m}_5 \quad \dots (i)$$

$$\text{energy balance} = \dot{m}_1 h_1 + \dot{m}_4 h_4 = \dot{m}_5 h_5 \quad \dots (ii)$$

From (i) and (ii)

$$\frac{\dot{m}_4}{\dot{m}_5} = \left(\frac{h_5 - h_1}{h_4 - h_1} \right)$$

Condenser as C.V

$$\dot{Q}_{out} = \dot{m}_4 (h_2 - h_3)$$

Boiler and two reheater as C.V respectively,

$$\dot{Q}_{in} = \dot{m}_5 (h_9 - h_6) + \dot{m}_5 (h_{11} - h_{10}) + \dot{m}_5 (h_1 - h_{12})$$

$$\eta = 1 - \frac{Q_{out}}{Q_{in}}$$

$$\eta = 1 - \frac{\dot{m}_4(h_2 - h_3)}{m_5(h_9 - h_6 + h_{11} - h_{10} + h_1 - h_{12})}$$

$$\text{Net work Output} = (\dot{Q}_{in} - \dot{Q}_{out}) \text{ or } (\eta \times \dot{Q}_{in})$$

Matlab code to calculate efficiency and quality for 2 reheater and 1 regeneration

```

clc;
clear all;
%%
h = Solution('liquidvapor.cti','water');
%% State 3
P3 = 10E+3;
setState_Psat(h,[P3,0]);
h3 = enthalpy_mass(h);
s3 = entropy_mass(h);
%% State 4
P4 = 0.8E+6; % Vary
s4 = s3;
setState_SP(h,[s4,P4]);
h4 = enthalpy_mass(h);
%% State 5
P5 = P4;
setState_Psat(h,[P5,0]);
h5 = enthalpy_mass(h);
s5 = entropy_mass(h);
%% State 6
P6 = 15E+6;
s6 = s5;
setState_SP(h,[s6,P6]);
h6 = enthalpy_mass(h);
%% State 7

```

```

P7 = P6;
setState_Psat(h,[P7,0]);
h7 = enthalpy_mass(h);
s7 = entropy_mass(h);
%% State 8
P8 = P6;
setState_Psat(h,[P8,1]);
h8 = enthalpy_mass(h);
%% State 9
P9 = P6;
T9 = 500+273.15;
set(h,'P',P9,'T',T9);
h9 = enthalpy_mass(h);
s9 = entropy_mass(h);
%% State 10
P10 = 7E+6; % vary
s10 = s9;
setState_SP(h,[s10,P10]);
h10 = enthalpy_mass(h);
%% State 11
P11 = P10;
T11 = 500+273.15;
set(h,'P',P11,'T',T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);
%% State 12
P12 = 3E+6; % vary
s12 = s11;
setState_SP(h,[s12,P12]);
h12 = enthalpy_mass(h);
%% State 1
P1 = P12;
T1 = 500+273.15;
set(h,'P',P1,'T',T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);
%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h,[s1_,P1_]);
h1_ = enthalpy_mass(h);
%% State 2
P2 = P3;
s2 = s1;
setState_SP(h,[s2,P2]);
h2 = enthalpy_mass(h);
setState_Psat(h,[P2,0]);
hf = enthalpy_mass(h);
setState_Psat(h,[P2,1]);
hg = enthalpy_mass(h);
hfg = hg-hf;
x = (h2-hf)/hfg;
m4_5 = ((h1_-h5)/(h1_-h4));
qout = m4_5*(h2-h3);
qin = h9-h6+h11-h10+h1-h12;
efficiency = 1-(qout/qin);

```

```

Wnet = qin-qout;
%% Printing the final results
fprintf("The efficiency of this cycle is %.4f %%\n", efficiency*100);
fprintf("The quality at condenser inlet is %.4f %%\n", x*100);

```

We found -

```

The efficiency of this cycle is 46.4560 %
The quality at condenser inlet is 87.7788 %

```

Thus, the Rankine cycle, featuring two reheaters and one regeneration, not only yields steam quality surpassing 85% at the condenser inlet but also achieves an efficiency exceeding 46%.

In exploring the reheat-regenerative configuration described in the above case (i.e. For two reheater and one regeneration), we systematically vary the boiler pressure (within the range of 12 MPa to 15 MPa) and the condenser pressure (within the range of 5 kPa to 10 kPa). By doing so, we construct a comprehensive plot to visually depict the influence of these pressure variations on both the thermal efficiency and net work output of the modified cycle.

Here is matlab code for the same

```

clc;
clear all;
%% Constants
Pb_range = linspace(12E+6, 15E+6, 30); % Range of boiler pressures from
12 MPa to 15 MPa
Pc_range = linspace(5E+3, 10E+3, 30); % Range of condenser pressures
from 5 kPa to 10 kPa
%% Initialize arrays to store results
efficiency_results = zeros(length(Pc_range), length(Pb_range));
Wnet_results = zeros(length(Pc_range), length(Pb_range));
%% Loop through different values of Pb and Pc
for i = 1:length(Pb_range)
    for j = 1:length(Pc_range)
        % Set boiler and condenser pressures
        P6 = Pb_range(i);
        P3 = Pc_range(j);
        % Call the existing code to calculate efficiency and quality (x)
    end
end

```

```

        [efficiency, Wnet] = calculate_cycle_performance(P6, P3);
        % Store results in arrays
        efficiency_results(j, i) = efficiency;
        Wnet_results(j, i) = Wnet;
    end
end
%% Create plots
figure;
% Plot for thermal efficiency
contourf(Pc_range, Pb_range, efficiency_results', 20, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
title('Effect of Boiler and Condenser Pressures on Thermal Efficiency');
figure;
% Plot for net Work output (Wnet)
contourf(Pc_range, Pb_range, Wnet_results', 20, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
title('Effect of Boiler and Condenser Pressures on net Work output (Wnet)');
% Define a function to calculate cycle performance
function [efficiency, Wnet] = calculate_cycle_performance(P6, P3)
    h = Solution('liquidvapor.cti', 'water');
    %% State 3
    setState_Psat(h, [P3, 0]);
    h3 = enthalpy_mass(h);
    s3 = entropy_mass(h);

    %% State 4
    P4 = 0.8E+6; % Vary
    s4 = s3;
    setState_SP(h, [s4, P4]);
    h4 = enthalpy_mass(h);

    %% State 5
    P5 = P4;
    setState_Psat(h, [P5, 0]);
    h5 = enthalpy_mass(h);
    s5 = entropy_mass(h);

    %% State 6
    s6 = s5;
    setState_SP(h, [s6, P6]);
    h6 = enthalpy_mass(h);

    %% State 7
    P7 = P6;
    setState_Psat(h, [P7, 0]);
    h7 = enthalpy_mass(h);

    %% State 8
    P8 = P6;
    setState_Psat(h, [P8, 1]);
    h8 = enthalpy_mass(h);

```

```

%% State 9
P9 = P6;
T9 = 500+273.15;
set(h, 'P', P9, 'T', T9);
h9 = enthalpy_mass(h);
s9 = entropy_mass(h);

%% State 10
P10 = 7E+6; % vary
s10 = s9;
setState_SP(h, [s10, P10]);
h10 = enthalpy_mass(h);

%% State 11
P11 = P10;
T11 = 500+273.15;
set(h, 'P', P11, 'T', T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);

%% State 12
P12 = 3E+6; % vary
s12 = s11;
setState_SP(h, [s12, P12]);
h12 = enthalpy_mass(h);

%% State 1
P1 = P12;
T1 = 500+273.15;
set(h, 'P', P1, 'T', T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);

%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h, [s1_, P1_]);
h1_ = enthalpy_mass(h);

%% State 2
P2 = P3;
s2 = s1;
setState_SP(h, [s2, P2]);
h2 = enthalpy_mass(h);

setState_Psat(h, [P2, 0]);
hf = enthalpy_mass(h);

setState_Psat(h, [P2, 1]);
hg = enthalpy_mass(h);

hfg = hg-hf;

x = (h2-hf)/hfg;

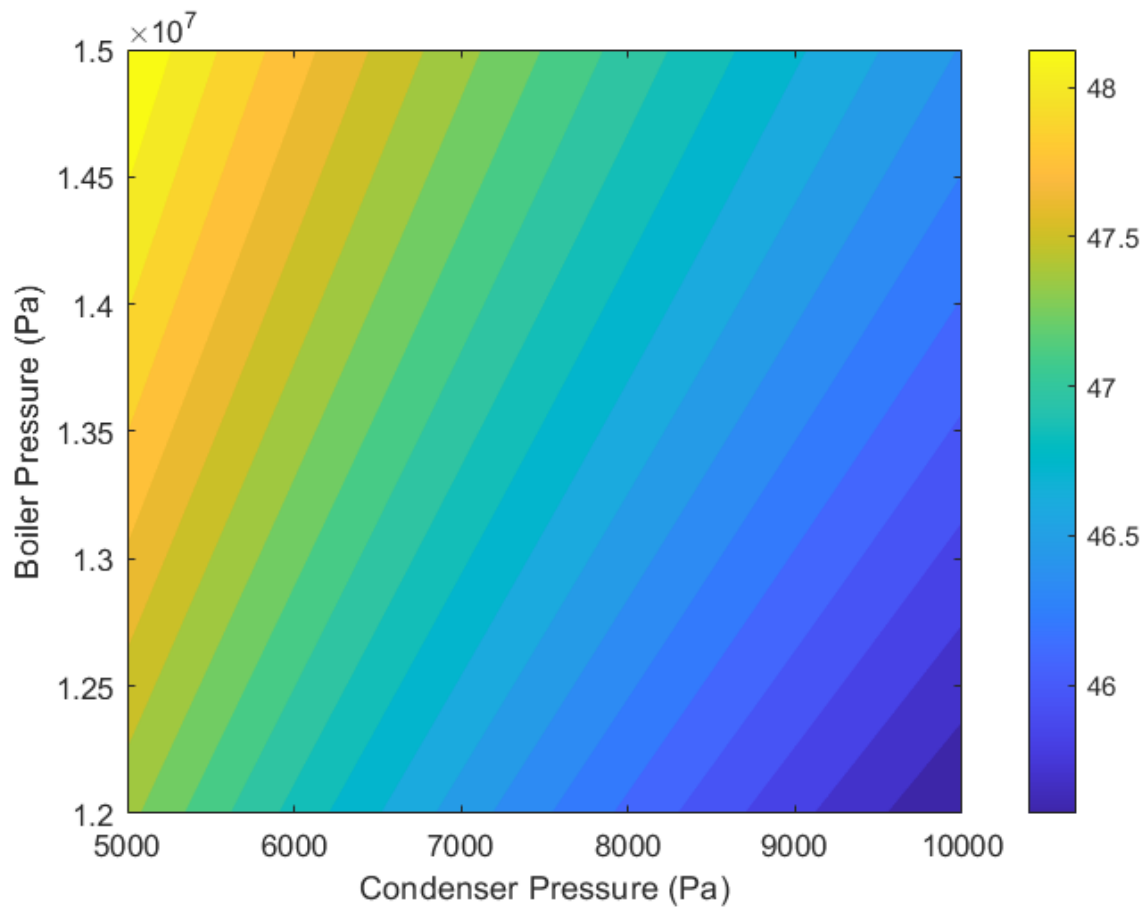
m4_5 = ((h1_-h5)/(h1_-h4));

```

```
qout = m4_5*(h2-h3);  
qin = h9-h6+h11-h10+h1-h12;  
  
efficiency = (1-(qout/qin))*100;  
  
Wnet = qin-qout;  
end
```

Results

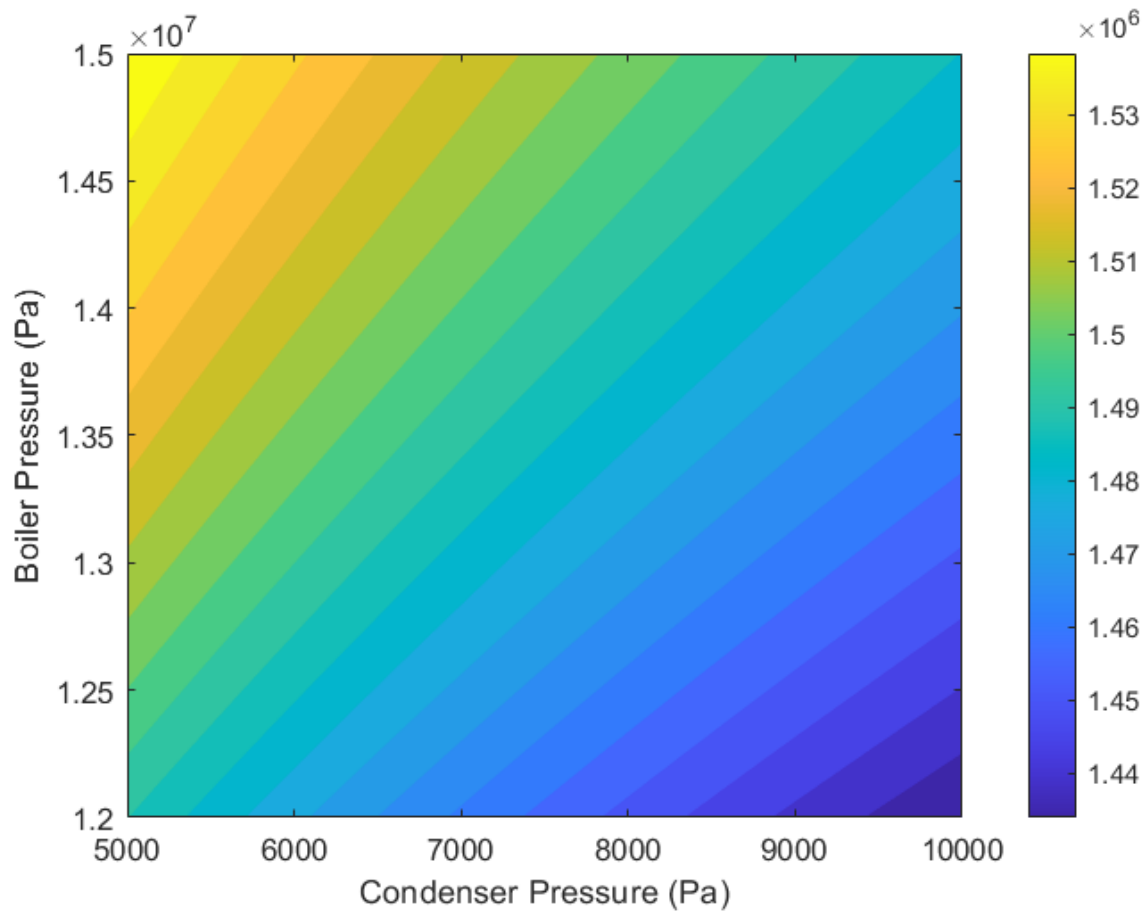
Effects of Boiler and Condenser Pressure on Efficiency



→ Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in thermal efficiency.

- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the thermal efficiency.

Effect of Boiler and Condenser Pressure on Net Work Output

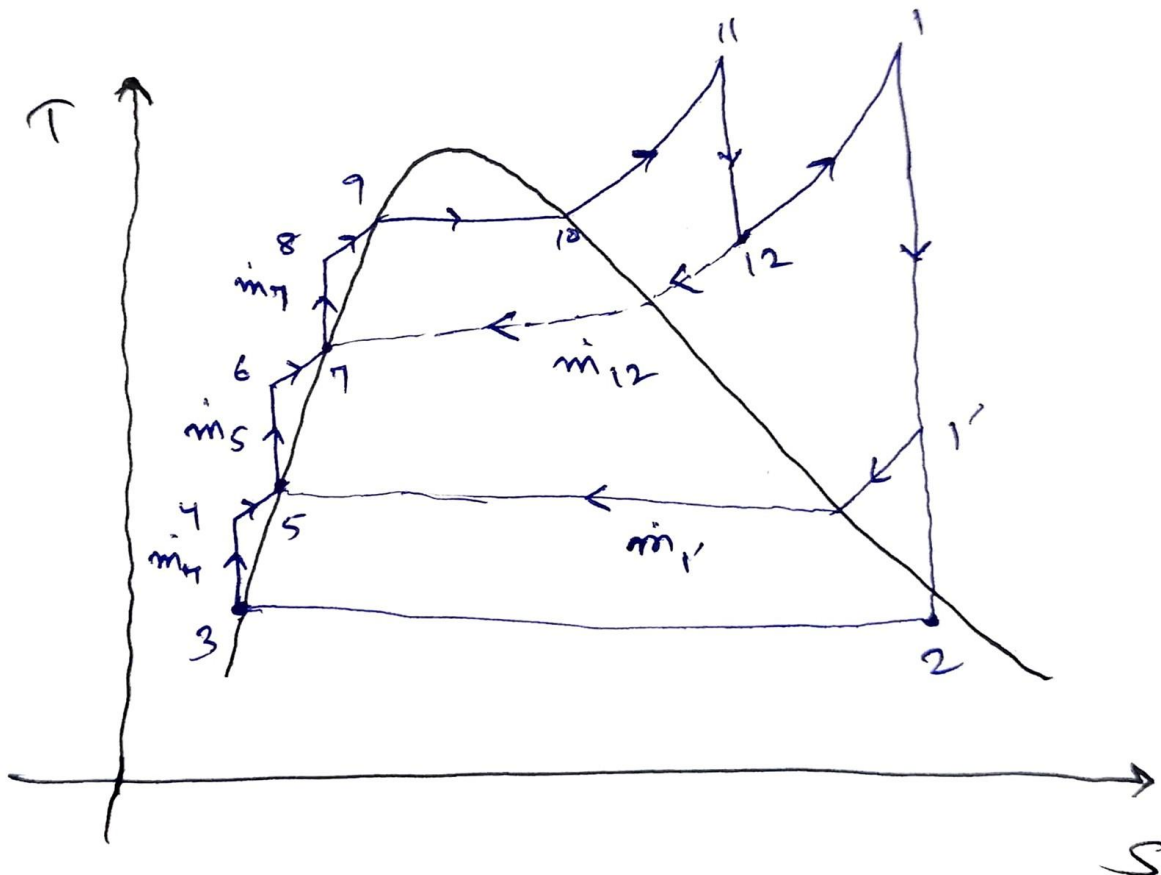


- Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in net work output.
- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the net work output.

4. Modified Ideal Rankine Cycle with one reheater and 2 regeneration

We also experimented with this combination out of curiosity to ascertain whether it aligns with the specified requirements.

The Rankine cycle with one reheater and two regenerators is an advanced configuration designed to maximize efficiency by incorporating both reheating and multiple stages of regeneration. Here's a brief overview of the processes that occur in the Rankine cycle with one reheater and two regenerators:



→ Isentropic Compression (Process 3-8):

- ◆ The working fluid, typically water or steam, is compressed in a pump from low pressure to high pressure.
- ◆ This compression process is usually isentropic, occurring with constant entropy.

→ Isobaric Heating in Boiler (Process 8-11):

- ◆ The high-pressure fluid is then heated in the boiler at constant pressure.
- ◆ Heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in temperature and volume.
- ◆ Steam leaves the boiler at a high temperature and pressure.
- Isentropic Expansion in High-Pressure Turbine (Process 11-12):
 - ◆ The high-pressure, high-temperature steam expands through the high-pressure turbine, converting some of its internal energy into mechanical work.
 - ◆ The steam exits the high-pressure turbine at an intermediate pressure and temperature.
- First Regeneration (Process 12-7):
 - ◆ Some of the steam from the intermediate-pressure turbine is extracted and sent to the first open feedwater heater.
 - ◆ The extracted steam transfers heat to the feedwater, increasing its temperature.
 - ◆ This process improves the overall efficiency by utilizing waste heat to preheat the feedwater.
- Reheat (Process 12-1):
 - ◆ The remaining steam after first regeneration is then sent to the reheater, where it is heated at constant pressure.
 - ◆ Reheating the steam improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Intermediate-Pressure Turbine (Process 12-1'):
 - ◆ The reheated steam expands through an intermediate-pressure turbine, further converting its internal energy into mechanical work.
 - ◆ The steam exits the intermediate-pressure turbine at a lower pressure and temperature.
- Second Regeneration (Process 1'-5):
 - ◆ Another portion of the steam from the intermediate-pressure turbine is extracted and sent to the second open feedwater heater.
 - ◆ The extracted steam transfers heat to the feedwater again, further increasing its temperature.
- Isentropic Expansion in Low-Pressure Turbine (Process 1'-2):
 - ◆ The twice-regenerated steam expands through the low-pressure turbine, converting its internal energy into mechanical work.
 - ◆ The steam exits the low-pressure turbine and is directed to the condenser.
- Isobaric Cooling in Condenser (Process 2-3):

- ◆ The low-pressure steam is condensed back into a liquid in the condenser, releasing heat to the surroundings.
- ◆ The condensed liquid is pumped back to the high-pressure state, completing the cycle.

This configuration optimizes the Rankine cycle by combining reheating with two stages of regeneration, utilizing waste heat more effectively to increase overall efficiency. The specific details of the cycle, such as pressures, temperatures, and the number of turbine stages, can vary depending on the design and requirements of the power plant.

Calculations

$$\text{mass balance} \Rightarrow \dot{m}_4 + \dot{m}_1 = \dot{m}_5 \quad \dots (i)$$

$$\text{energy balance} \Rightarrow \dot{m}_4 h_4 + \dot{m}_1 h_1 = \dot{m}_5 h_5 \quad \dots (ii)$$

From (i) & (ii)

$$\frac{\dot{m}_4}{\dot{m}_5} = \frac{h_5 - h_1'}{h_4 - h_1'} \quad \dots (iii)$$

Now,

At state (7), Second open feedwater heater as CV

$$\text{mass balance} \Rightarrow \dot{m}_5 + \dot{m}_{12} = \dot{m}_7 \quad \dots (iv)$$

$$\text{energy balance} \Rightarrow \dot{m}_5 h_6 + \dot{m}_{12} h_{12} = \dot{m}_7 h_7 \quad \dots (v)$$

From (iv) and (v),

$$\frac{\dot{m}_7}{\dot{m}_5} = \frac{h_6 - h_{12}}{h_7 - h_{12}} \quad \dots (vi)$$

Condenser as CV

$$\dot{Q}_{out} = \dot{m}_4(h_2 - h_3)$$

Boiler as CV: $\dot{Q}_b = \dot{m}_n(h_{11} - h_8)$

Reheater as CV: $\dot{Q}_r = \dot{m}_5(h_1 - h_{12})$

$$\dot{Q}_{in} = \dot{m}_5(h_1 - h_{12}) + \dot{m}_n(h_{11} - h_8)$$

Now, Efficiency

$$\eta = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{\dot{m}_4(h_2 - h_3)}{\dot{m}_5(h_1 - h_{12}) + \dot{m}_n(h_{11} - h_8)}$$

$$\text{Net work Output} = (\dot{Q}_{in} - \dot{Q}_{out}) \text{ or } (\eta \times \dot{Q}_{in})$$

Matlab code to calculate the efficiency and quality for 1 reheater and 2 regeneration

```
clc;
clear all;
%%% ONE REHEATER AND 2 REGENERATION
%%
h = Solution('liquidvapor.cti','water');
%% State 3
P3 = 10E+3;
setState_Psat(h, [P3,0]);
h3 = enthalpy_mass(h);
s3 = entropy_mass(h);
%% State 4
P4 = 0.8E+6; % Vary
s4 = s3;
setState_SP(h, [s4,P4]);
h4 = enthalpy_mass(h);
%% State 5
P5 = P4;
setState_Psat(h, [P5,0]);
h5 = enthalpy_mass(h);
s5 = entropy_mass(h);
%% State 6
P6 = 3.7E+6; % vary
s6 = s5;
setState_SP(h, [s6,P6]);
h6 = enthalpy_mass(h);
%% State 7
P7 = P6;
setState_Psat(h, [P7,0]);
h7 = enthalpy_mass(h);
s7 = entropy_mass(h);
%% State 8
P8 = 15E+6;
s8 = s7;
setState_SP(h, [s8,P8]);
h8 = enthalpy_mass(h);
%% State 9
P9 = P8;
setState_Psat(h, [P9,0]);
h9 = enthalpy_mass(h);
%% State 10
P10 = P8;
setState_Psat(h, [P10,1]);
h10 = enthalpy_mass(h);
%% State 11
P11 = P8;
T11 = 500+273.15;
set(h, 'P', P11, 'T', T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);
%% State 12
P12 = P6;
s12 = s11;
setState_SP(h, [s12,P12]);
```

```

h12 = enthalpy_mass(h);
%% State 1
P1 = P6;
T1 = 500+273.15;
set(h, 'P', P1, 'T', T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);
%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h, [s1_, P1_]);
h1_ = enthalpy_mass(h);
%% State 2
P2 = P3;
s2 = s1;
setState_SP(h, [s2, P2]);
h2 = enthalpy_mass(h);
setState_Psat(h, [P2, 0]);
hf = enthalpy_mass(h);
setState_Psat(h, [P2, 1]);
hg = enthalpy_mass(h);
hfg = hg-hf;
x = ((h2-hf)/hfg)*100;
m4_6 = ((h1_-h5)/(h1_-h4));
m7_6 = ((h6-h12)/(h7-h12));
qout = m4_6*(h2-h3);
qin = (m7_6*(h11-h8))+h1-h12;
efficiency = (1-(qout/qin))*100;
Wnet = qin-qout;
fprintf("The quality of the modified cycle is %.4f %%\n", x);
fprintf("The Efficiency of the modified cycle is %.4f %%\n", efficiency);

```

We found -

```

The quality of the modified cycle is 86.3861 %
The Efficiency of the modified cycle is 46.7726 %

```

Thus, a Rankine cycle with one reheater and two regeneration can also give efficiency more than 46% and quality (at the condenser inlet) above 85%.

Now, we systematically vary both boiler and condenser pressures and generate a plot to elucidate the impact of these pressure changes on thermal efficiency and net work output.

Here is the matlab code for the same

```

clc;
clear all;
%% Constants
Pb_range = linspace(12E+6, 15E+6, 20); % Range of boiler pressures from 12 MPa
to 15 MPa
Pc_range = linspace(5E+3, 10E+3, 20); % Range of condenser pressures from 5
kPa to 10 kPa
%% Initialize arrays to store results
efficiency_results = zeros(length(Pc_range), length(Pb_range));
Wnet_results = zeros(length(Pc_range), length(Pb_range));
%% Loop through different values of Pb and Pc
for i = 1:length(Pb_range)
    for j = 1:length(Pc_range)
        % Set boiler and condenser pressures
        P8 = Pb_range(i);
        P3 = Pc_range(j);
        % Call the existing code to calculate efficiency and quality (x)
        [efficiency, Wnet] = calculate_cycle_performance(P3, P8);
        % Store results in arrays
        efficiency_results(j, i) = efficiency;
        Wnet_results(j, i) = Wnet;
    end
end
%% Create plots
figure;
% Plot for thermal efficiency
contourf(Pc_range, Pb_range, efficiency_results', 20, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
title('Effect of Boiler and Condenser Pressures on Thermal Efficiency');
figure;
% Plot for net Work output (Wnet)
contourf(Pc_range, Pb_range, Wnet_results', 30, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
title('Effect of Boiler and Condenser Pressures on net work output (x)');
%% Define a function to calculate cycle performance
function [efficiency, Wnet] = calculate_cycle_performance(P3, P8)
    h = Solution('liquidvapor.cti', 'water');
    %% State 3
    setState_Psat(h, [P3, 0]);
    h3 = enthalpy_mass(h);
    s3 = entropy_mass(h);

    %% State 4
    P4 = 0.8E+6; % Vary

```

```

s4 = s3;
setState_SP(h, [s4,P4]);
h4 = enthalpy_mass(h);

%% State 5
P5 = P4;
setState_Psat(h, [P5,0]);
h5 = enthalpy_mass(h);
s5 = entropy_mass(h);

%% State 6
P6 = 3.7E+6;                % vary
s6 = s5;
setState_SP(h, [s6,P6]);
h6 = enthalpy_mass(h);

%% State 7
P7 = P6;
setState_Psat(h, [P7,0]);
h7 = enthalpy_mass(h);
s7 = entropy_mass(h);

%% State 8
s8 = s7;
setState_SP(h, [s8,P8]);
h8 = enthalpy_mass(h);

%% State 9
P9 = P8;
setState_Psat(h, [P9,0]);
h9 = enthalpy_mass(h);

%% State 10
P10 = P8;
setState_Psat(h, [P10,1]);
h10 = enthalpy_mass(h);

%% State 11
P11 = P8;
T11 = 500+273.15;
set(h, 'P', P11, 'T', T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);

%% State 12
P12 = P6;
s12 = s11;
setState_SP(h, [s12,P12]);
h12 = enthalpy_mass(h);

%% State 1
P1 = P6;
T1 = 500+273.15;
set(h, 'P', P1, 'T', T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);

```



```

%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h, [s1_, P1_]);
h1_ = enthalpy_mass(h);

%% State 2
P2 = P3;
s2 = s1;
setState_SP(h, [s2, P2]);
h2 = enthalpy_mass(h);

setState_Psat(h, [P2, 0]);
hf = enthalpy_mass(h);

setState_Psat(h, [P2, 1]);
hg = enthalpy_mass(h);

hfg = hg-hf;

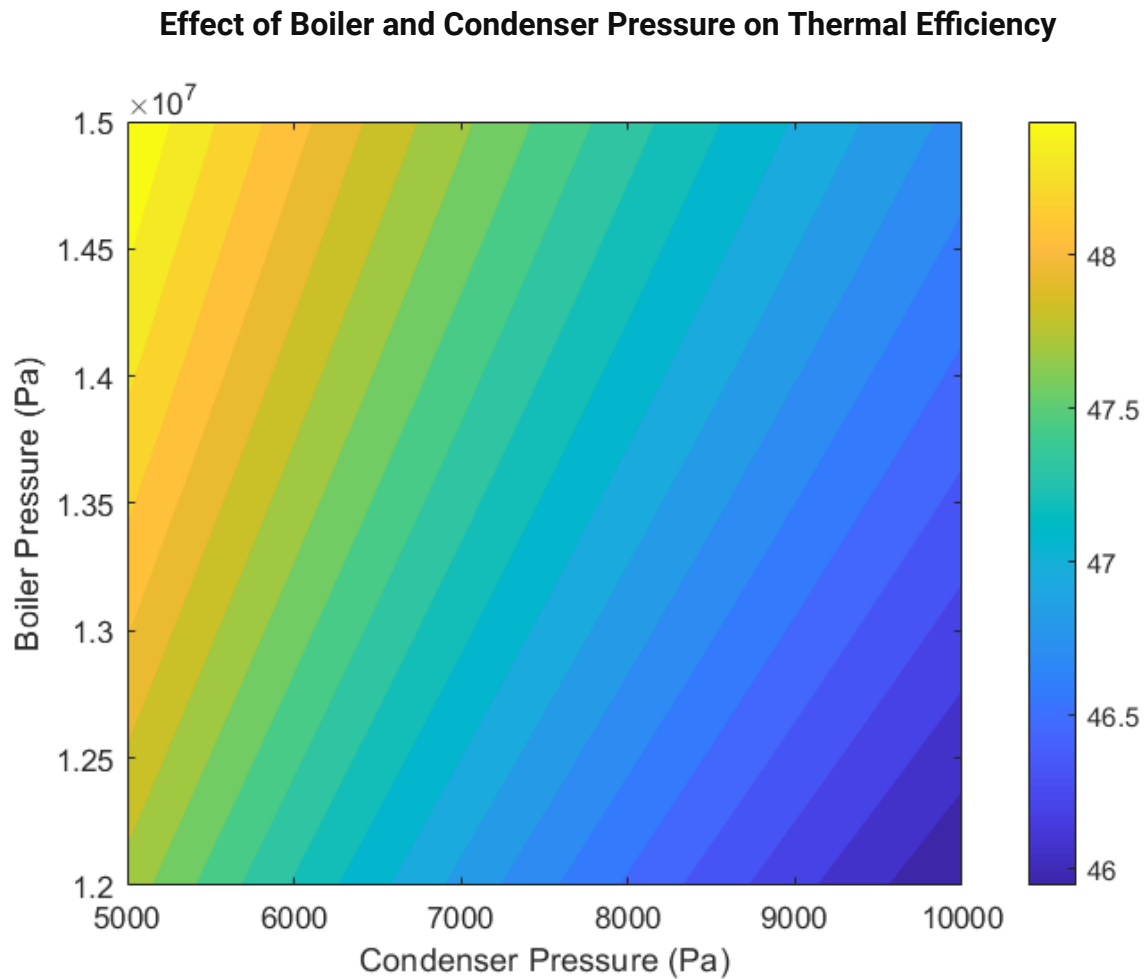
x = ((h2-hf)/hfg);

m4_6 = ((h1_-h5)/(h1_-h4));
m7_6 = ((h6-h12)/(h7-h12));
qout = m4_6*(h2-h3);
qin = (m7_6*(h11-h8))+h1-h12;
Wnet = qin-qout;

efficiency = (1-(qout/qin))*100;
end

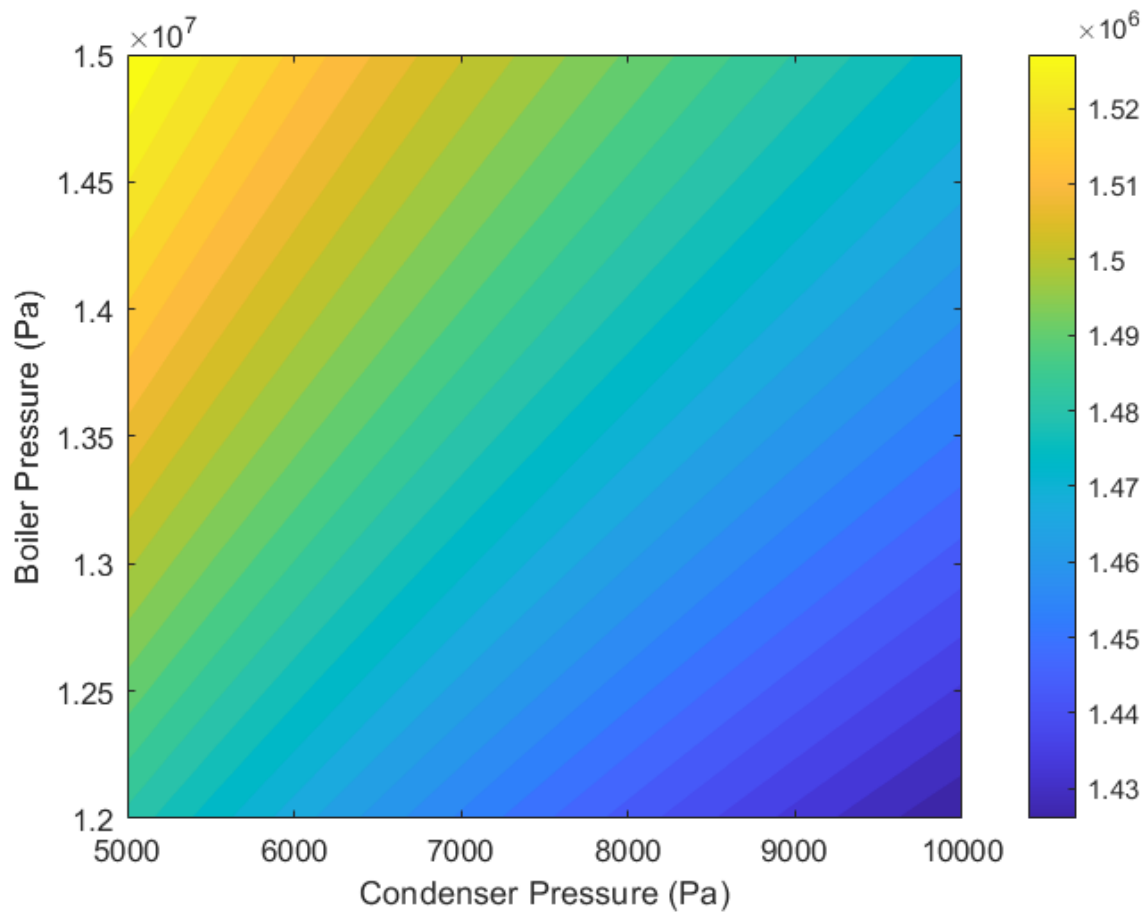
```

Results



- Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in thermal efficiency.
- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the thermal efficiency.

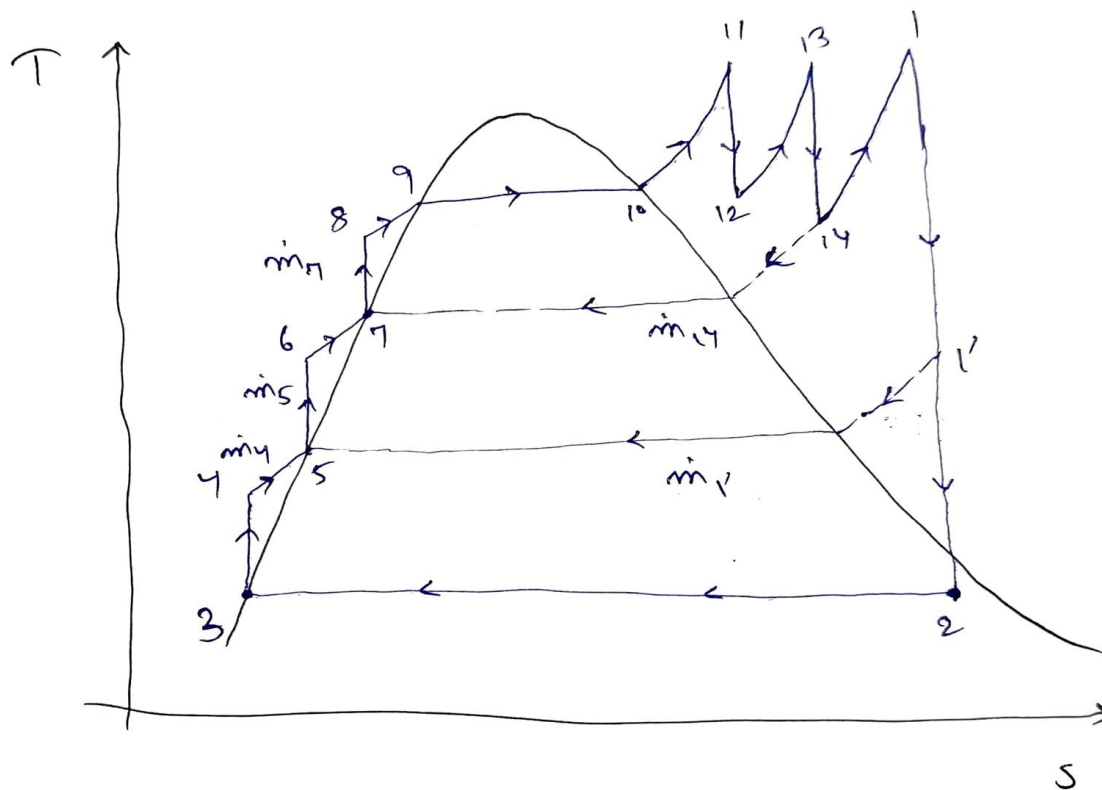
Effect of Boiler and Condenser Pressure on Net Work Output



- Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in net work output.
- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the net work output.

5. Modified Ideal Rankine Cycle with 2 reheater and 2 regeneration

The Rankine cycle with two reheaters and two regenerators is an advanced and highly efficient configuration designed to optimize the power generation process by incorporating both reheating and multiple stages of regeneration. Here's a brief overview of the processes that occur in the Rankine cycle with two reheaters and two regenerators:



→ Isentropic Compression (Process 3-8):

- ◆ The working fluid, typically water or steam, is compressed in a pump from low pressure to high pressure.
- ◆ This compression process is generally isentropic, occurring with constant entropy.

→ Isobaric Heating in Boiler (Process 8-11):

- ◆ The high-pressure fluid is heated in the first stage of the boiler at constant pressure.

- ◆ Heat input causes the fluid to undergo phase change (from liquid to vapor in the case of water) and results in a significant increase in temperature and volume.
- ◆ Steam leaves the first stage of the boiler at a high temperature and pressure.
- Isentropic Expansion in High-Pressure Turbine (Process 11-12):
 - ◆ The high-pressure, high-temperature steam expands through the high-pressure turbine, converting some of its internal energy into mechanical work.
 - ◆ The steam exits the high-pressure turbine at an intermediate pressure and temperature.
- Reheat 1 (Process 12-13):
 - ◆ The intermediate-pressure steam is extracted from the high-pressure turbine and sent to the first reheater, where it is heated at constant pressure.
 - ◆ Reheating the steam improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Intermediate-Pressure Turbine (Process 13-14):
 - ◆ The reheated steam expands through an intermediate-pressure turbine, further converting its internal energy into mechanical work.
 - ◆ The steam exits the intermediate-pressure turbine at a lower pressure and temperature.
- First Regeneration (Process 14-7):
 - ◆ Some of the steam from the second reheater is extracted and sent to the first regenerator.
 - ◆ The extracted steam transfers heat to the feedwater, increasing its temperature before it enters the first stage of the boiler.
 - ◆ This process improves the overall efficiency by utilizing waste heat to preheat the feedwater.
- Reheat 2 (Process 14-1):
 - ◆ The low-pressure steam is extracted from the intermediate-pressure turbine and sent to the second reheater, where it undergoes further reheating at constant pressure.
 - ◆ Reheating the steam again improves its quality by increasing its temperature without raising its pressure.
- Isentropic Expansion in Low-Pressure Turbine (Process 1-2):
 - ◆ The twice-reheated steam expands through the low-pressure turbine, converting its internal energy into mechanical work.
 - ◆ The steam exits the low-pressure turbine and is directed to the condenser.

→ Second Regeneration (Process 1'-5):

- ◆ Another portion of the steam from the low pressure turbine is extracted and sent to the second regenerator.
- ◆ The extracted steam transfers heat to the feedwater again, further increasing its temperature before it enters the second stage of the boiler.

→ Isobaric Cooling in Condenser (Process 2-3):

The low-pressure steam is condensed back into a liquid in the condenser, releasing heat to the surroundings.

The condensed liquid is pumped back to the high-pressure state, completing the cycle.

This advanced configuration optimizes the Rankine cycle by combining two stages of reheating with two stages of regeneration, allowing for efficient utilization of waste heat and increasing overall cycle efficiency. The specific details of the cycle, such as pressures, temperatures, and the number of turbine stages, can vary depending on the design and requirements of the power plant.

Calculation:

At state (5) First open feed water heater as CV:

$$\text{mass balance} \Rightarrow \dot{m}_1 + \dot{m}_4 = \dot{m}_5 \quad \dots (i)$$

$$\text{energy balance} \Rightarrow \dot{m}_4 h_4 + \dot{m}_1 h_1 = \dot{m}_5 h_5 \quad \dots (ii)$$

From (i) & (ii)

$$\frac{\dot{m}_4}{\dot{m}_5} = \frac{h_5 - h_1'}{h_4 - h_1'} \quad \dots (iii)$$

At state (7), second open feed water heater as CV,

$$\text{mass balance} \Rightarrow \dot{m}_5 + \dot{m}_{14} = \dot{m}_7 \quad \dots (iv)$$

$$\text{energy balance} \Rightarrow \dot{m}_5 h_6 + \dot{m}_{14} h_{14} = \dot{m}_7 h_7 \quad \dots (v)$$

From (iv) & (v)

$$\frac{\dot{m}_7}{\dot{m}_5} = \frac{h_6 - h_{14}}{h_7 - h_{14}} \quad \dots (vi)$$

From state (8) to (11),

$$\text{Boiler as CV:} \quad \dot{Q}_b = \dot{m}_7 (h_{11} - h_8)$$

$$\text{1st Reheater as CV:} \quad \dot{Q}_{r1} = \dot{m}_7 (h_{13} - h_{12})$$

$$\text{2nd Reheater as CV:} \quad \dot{Q}_{r2} = \dot{m}_5 (h_1 - h_{14})$$

$$\text{Total Heat Input: } \dot{Q}_{in} = \dot{Q}_b + \dot{Q}_{r1} + \dot{Q}_{r2}$$

$$\dot{Q}_{in} = \dot{m}_7 (h_{11} - h_8) + \dot{m}_7 (h_{13} - h_{12}) + \dot{m}_5 (h_1 - h_{14})$$

Heat Output: $\dot{Q}_{out} = \dot{m}_4(h_2 - h_3)$

Now, Efficiency

$$\eta = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{\dot{m}_{4.5}(h_2 - h_3)}{\dot{m}_{7.5}(h_{11} - h_8 + h_{13} - h_{12}) + (h_1 - h_{12})}$$

$$\eta = 1 - \frac{\dot{m}_{4.5}(h_2 - h_3)}{\dot{m}_{7.5}(h_{11} - h_8 + h_{13} - h_{12}) + (h_1 - h_{12})}$$

Matlab code to calculate the efficiency and quality for 2 reheater and 2 regeneration

```
clc;
clear all;
%2reh_2reg
%%
h = Solution('liquidvapor.cti','water');
%% State 3
P3 = 10E+3;
setState_Psat(h, [P3,0]);
h3 = enthalpy_mass(h);
s3 = entropy_mass(h);
%% State 4
P4 = 0.05E+6; % Vary
s4 = s3;
setState_SP(h, [s4,P4]);
h4 = enthalpy_mass(h);
%% State 5
P5 = P4;
setState_Psat(h, [P5,0]);
h5 = enthalpy_mass(h);
s5 = entropy_mass(h);
%% State 6
```



```

P6 = 2.1E+6;                % vary
s6 = s5;
setState_SP(h, [s6, P6]);
h6 = enthalpy_mass(h);
%% State 7
P7 = P6;
setState_Psat(h, [P7, 0]);
h7 = enthalpy_mass(h);
s7 = entropy_mass(h);
%% State 8
P8 = 15E+6;
s8 = s7;
setState_SP(h, [s8, P8]);
h8 = enthalpy_mass(h);
%% State 9
P9 = P8;
setState_Psat(h, [P9, 0]);
h9 = enthalpy_mass(h);
%% State 10
P10 = P8;
setState_Psat(h, [P10, 1]);
h10 = enthalpy_mass(h);
%% State 11
P11 = P8;
T11 = 500+273.15;
set(h, 'P', P11, 'T', T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);
%% State 12
P12 = 1.8E+6;                % vary
s12 = s11;
setState_SP(h, [s12, P12]);
h12 = enthalpy_mass(h);
%% State 13
P13 = P12;
T13 = 500+273.15;
set(h, 'P', P13, 'T', T13);
h13 = enthalpy_mass(h);
s13 = entropy_mass(h);
%% State 14
P14 = P6;
s14 = s13;
setState_SP(h, [s14, P14]);
h14 = enthalpy_mass(h);
%% State 1
P1 = P6;
T1 = 500+273.15;
set(h, 'P', P1, 'T', T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);
%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h, [s1_, P1_]);
h1_ = enthalpy_mass(h);
%% State 2

```

```

P2 = P3;
s2 = s1;
setState_SP(h,[s2,P2]);
h2 = enthalpy_mass(h);
setState_Psat(h,[P2,0]);
hf = enthalpy_mass(h);
setState_Psat(h,[P2,1]);
hg = enthalpy_mass(h);
hfg = hg-hf;
x = ((h2-hf)/hfg)*100;
m4_5 = ((h1_-h5)/(h1_-h4));
m7_5 = ((h6-h12)/(h7-h12));
qout = m4_5*( h2 - h3 );
qin = ( m7_5 * (h11-h8+h13-h12) )+h1-h14;
efficiency = (1-(qout/qin))*100;
fprintf("The quality of the modified rankine cycle is %.4f %%\n", x);
fprintf("The efficiency of the modified rankine cycle is %.4f %%\n",
efficiency);

```

We found-

```

The quality of the modified rankine cycle is 90.1028 %
The efficiency of the modified rankine cycle is 48.8825 %

```

Thus, a Rankine cycle with two reheater and two regeneration gives efficiency more than 48% and quality (at the condenser inlet) above 90%.

Now, we systematically vary both boiler and condenser pressures and generate a plot to elucidate the impact of these pressure changes on thermal efficiency and net work output.

Here is Matlab code for the same

```

clc;
clear all;
%% Constants
Pb_range = linspace(12E+6, 15E+6, 20); % Range of boiler pressures from 12
MPa to 15 MPa
Pc_range = linspace(5E+3, 10E+3, 20); % Range of condenser pressures from 5
kPa to 10 kPa
%% Initialize arrays to store results
efficiency_results = zeros(length(Pc_range), length(Pb_range));
Wnet_results = zeros(length(Pc_range), length(Pb_range));
%% Loop through different values of Pb and Pc
for i = 1:length(Pb_range)
    for j = 1:length(Pc_range)
        % Set boiler and condenser pressures
        P8 = Pb_range(i);
        P3 = Pc_range(j);
    end
end

```

```

        % Call the existing code to calculate efficiency and quality (x)
        [efficiency, Wnet] = calculate_cycle_performance(P3, P8);
        % Store results in arrays
        efficiency_results(j, i) = efficiency;
        Wnet_results(j, i) = Wnet;
    end
end
%% Create plots
figure;
% Plot for thermal efficiency
contourf(Pc_range, Pb_range, efficiency_results', 20, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
%title('Effect of Boiler and Condenser Pressures on Thermal Efficiency');
figure;
% Plot for net Work output (Wnet)
contourf(Pc_range, Pb_range, Wnet_results', 30, 'LineColor', 'none');
colorbar;
xlabel('Condenser Pressure (Pa)');
ylabel('Boiler Pressure (Pa)');
%title('Effect of Boiler and Condenser Pressures on net work output (Wnet)');
%% Define a function to calculate cycle performance
function [efficiency, Wnet] = calculate_cycle_performance(P3, P8)
    h = Solution('liquidvapor.cti', 'water');
    %% State 3
    setState_Psat(h, [P3, 0]);
    h3 = enthalpy_mass(h);
    s3 = entropy_mass(h);

    %% State 4
    P4 = 0.05E+6; % Vary
    s4 = s3;
    setState_SP(h, [s4, P4]);
    h4 = enthalpy_mass(h);

    %% State 5
    P5 = P4;
    setState_Psat(h, [P5, 0]);
    h5 = enthalpy_mass(h);
    s5 = entropy_mass(h);

    %% State 6
    P6 = 2.1E+6; % vary
    s6 = s5;
    setState_SP(h, [s6, P6]);
    h6 = enthalpy_mass(h);

    %% State 7
    P7 = P6;
    setState_Psat(h, [P7, 0]);
    h7 = enthalpy_mass(h);
    s7 = entropy_mass(h);

    %% State 8

```

```

s8 = s7;
setState_SP(h,[s8,P8]);
h8 = enthalpy_mass(h);

%% State 9
P9 = P8;
setState_Psat(h,[P9,0]);
h9 = enthalpy_mass(h);

%% State 10
P10 = P8;
setState_Psat(h,[P10,1]);
h10 = enthalpy_mass(h);

%% State 11
P11 = P8;
T11 = 500+273.15;
set(h,'P',P11,'T',T11);
h11 = enthalpy_mass(h);
s11 = entropy_mass(h);

%% State 12
P12 = 1.8E+6;           % vary
s12 = s11;
setState_SP(h,[s12,P12]);
h12 = enthalpy_mass(h);

%% State 13
P13 = P12;
T13 = 500+273.15;
set(h,'P',P13,'T',T13);
h13 = enthalpy_mass(h);
s13 = entropy_mass(h);

%% State 14
P14 = P6;
s14 = s13;
setState_SP(h,[s14,P14]);
h14 = enthalpy_mass(h);

%% State 1
P1 = P6;
T1 = 500+273.15;
set(h,'P',P1,'T',T1);
h1 = enthalpy_mass(h);
s1 = entropy_mass(h);

%% State 1_
P1_ = P4;
s1_ = s1;
setState_SP(h,[s1_,P1_]);
h1_ = enthalpy_mass(h);

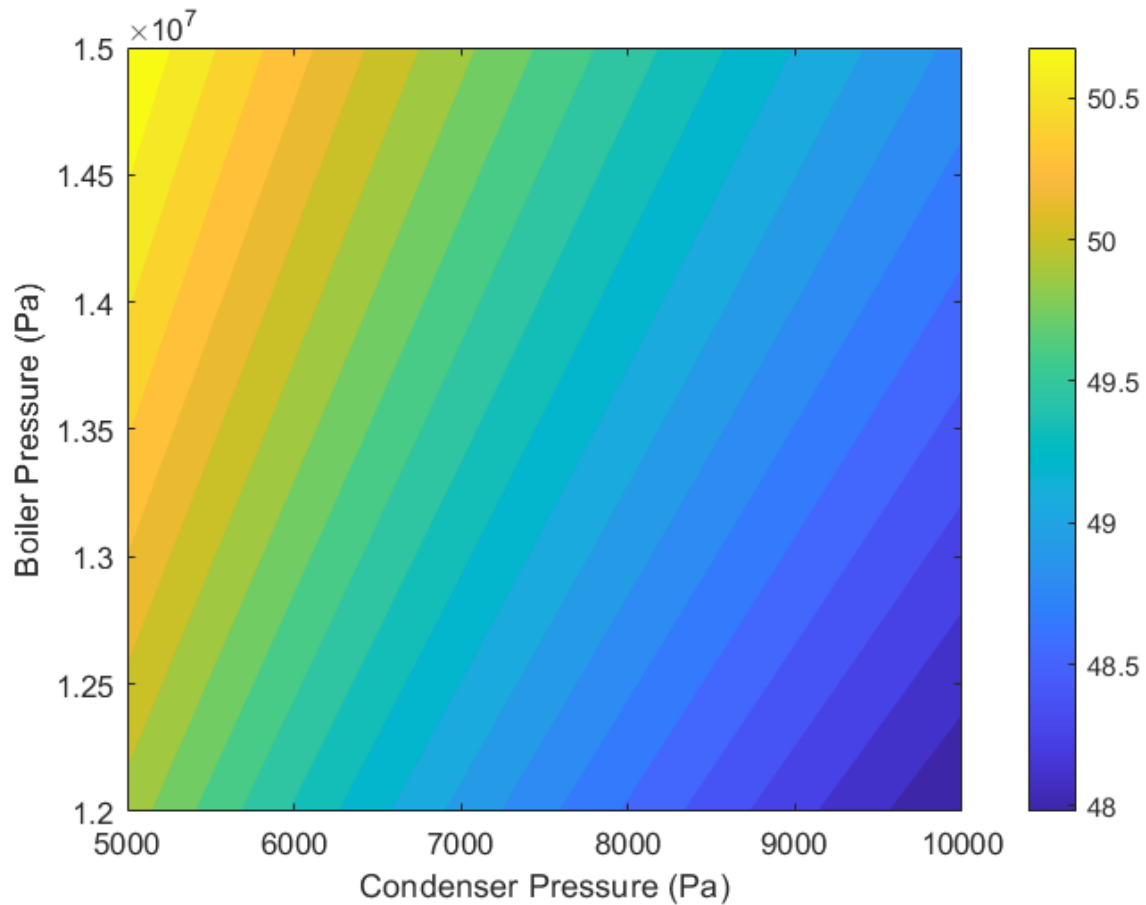
%% State 2
P2 = P3;
s2 = s1;

```

```
setState_SP(h,[s2,P2]);  
h2 = enthalpy_mass(h);  
  
setState_Psat(h,[P2,0]);  
hf = enthalpy_mass(h);  
  
setState_Psat(h,[P2,1]);  
hg = enthalpy_mass(h);  
  
hfg = hg-hf;  
  
x = ((h2-hf)/hfg);  
  
m4_5 = ((h1_-h5)/(h1_-h4));  
m7_5 = ((h6-h12)/(h7-h12));  
  
qout = m4_5*( h2 - h3 );  
qin = ( m7_5 * (h11-h8+h13-h12) )+h1-h14;  
  
efficiency = (1-(qout/qin));  
Wnet = efficiency*qin;  
end
```

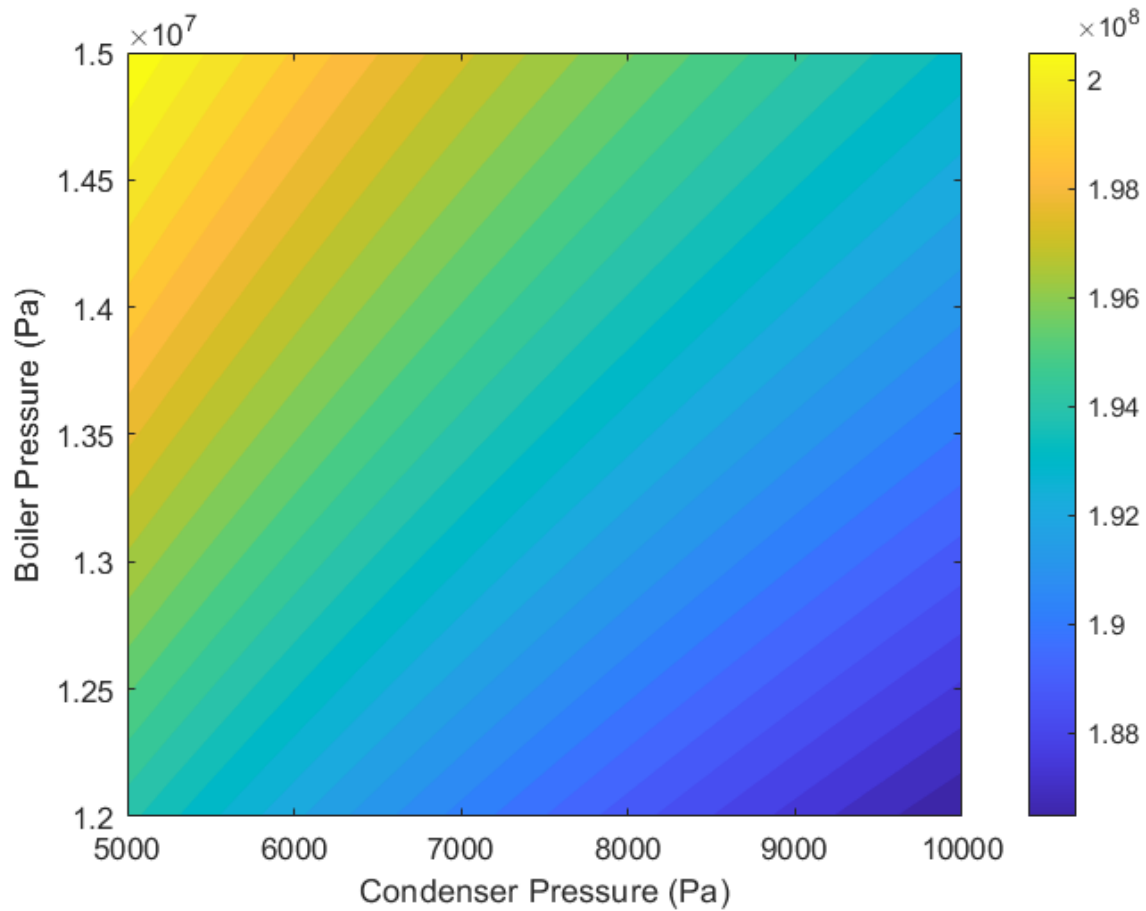
Results-

Effect of Boiler and Condenser Pressure on Thermal Efficiency



- Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in thermal efficiency.
- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the thermal efficiency.

Effect of Boiler and Condenser Pressure on Net Work Output



- Elevating the boiler pressure while maintaining a constant condenser pressure results in an increase in net work output.
- Increasing the condenser pressure while maintaining a constant boiler pressure results in decrease in the net work output.

Table:

Here is a refined comparison table showcasing the efficiency and quality variations among the aforementioned cases:

Cases	Efficiency (in %)	Quality (in %)
1 Reheater	42.8025	87.7788
1 Reheater and 1 Regeneration	45.6765	86.2073
2 Reheater and 1 Regeneration	46.4560	87.7788
1 Reheater and 2 Regeneration	46.7726	86.3861
2 Reheater and 2 Regeneration	48.8825	90.1028

Observations

During the trial of each aforementioned case, our observations indicate that introducing additional reheaters, while maintaining the boiler and condenser pressure constant as well as turbine temperature to 500 °C, leads to an improvement in the **quality** at the condenser inlet within the modified Rankine Cycle.

Similarly, when augmenting the regeneration process without altering the boiler and condenser pressure, and ensuring the turbine temperature remains to 500 °C, we observe an increase in the thermal efficiency of the modified Rankine cycle.